Table 1. Metadata for acoustical monitoring sites at Gates of the Arctic National Park and Preserve (GAAR). Dates reflect each site's recorded NVSPL deployment period(s).

Site	Site Name	Dates	Vegetation	Wilderness	Elevation (m)	Latitude	Longitude
GAARAMLA	NA		NA	NA	NA	NA	NA
GAARANAK	NA		NA	NA	NA	NA	NA
GAARARRI	NA		NA	NA	NA	NA	NA
GAARFLOR	NA		NA	NA	NA	NA	NA
GAARHAMM	NA		NA	NA	NA	NA	NA
GAARITKI	NA		NA	NA	NA	NA	NA
GAARKURU	NA		NA	NA	NA	NA	NA
GAARMWAL	NA		NA	NA	NA	NA	NA
GAARNARV	NA		NA	NA	NA	NA	NA
GAARNNUT	NA		NA	NA	NA	NA	NA
GAARNUTU	NA		NA	NA	NA	NA	NA
GAARNWAL	NA		NA	NA	NA	NA	NA
GAARPALA	NA		NA	NA	NA	NA	NA
GAARSELB	NA		NA	NA	NA	NA	NA
GAARUPNO	NA		NA	NA	NA	NA	NA
GAARWALK	NA		NA	NA	NA	NA	NA

Table 2. Summary of sound levels at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2019. For each site, includes the 90th percentile (LA10), existing (LA50), natural (LAnat), and 10th percentile (LA90) ambient sound levels (dB re 20 uPa, A−weighted broadband, 12.5−20,000 Hz) during the day (0700−1900) and night (1900−0700).

Time	SiteID	LA10	LA50	LAnat	LA90
Day	GAARITKI	29.5	25.5	25.5	23.2
Night	GAARITKI	31.9	28.5	28.5	26.4
Day	GAARPALA	27.4	21.0	20.9	18.3
Night	GAARPALA	25.6	19.5	19.5	17.4

Table 3. Number of days of audio analyzed for the Listening Center and SPLAT detection methods at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2019 − the basis for the analysis−period text reported in the Methods section.

Type	DaysText
ListeningCenter	
SPLAT	11−39

Table 4. Executive summary for Gates of the Arctic National Park and Preserve (GAAR), Summer 2019: mean overall noise audibility, top contributing noise sources, and ambient sound levels (LA50/LAnat, day and night) by site.

SiteID	Noise	Aircraft	NA 3	NA 4	LA50 Day	LAnat Day	LA50 Night	LAnat Night
GAARITKI	0.4	0.4	NA	NA	25.5	25.5	28.5	28.5
GAARPALA	0.8	0.8	NA	NA	21.0	20.9	19.5	19.5

Table 5. Noise impact (dB, existing minus natural ambient) and Listening Area Reduction (%) by site at Gates of the Arctic National Park and Preserve (GAAR), Summer 2019, for day, night, and the combined 24−hour period.

SiteID	DayImpact	DayLAR	NightImpact	NightLAR	AllImpact	AllLAR
GAARITKI	0.0	0.0	0	0	0.0	0.0
GAARPALA	0.1	2.3	0	0	0.1	1.1

Table 6. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2019.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARITKI	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.3	0.3	0.0	0.0	0.0
GAARITKI	1.1	1	Aircraft	1.1	Jet	1.0	0.0	1.0	172.8	0.0	172.8
GAARITKI	1.2	1	Aircraft	1.2	Prop	0.6	0.2	0.8	245.1	162.5	226.8
GAARITKI	1.3	1	Aircraft	1.3	Helicopter	0.2	0.0	0.2	106.5	0.0	106.5

Table 7. SPLAT: mean number of detected events and mean event length (seconds) for each sound source, by day, night, and the combined 24−hour period, in Summer 2019.

SiteID	SrcID	Type	Source Type	SrcID 1	Source Description	Day Count	Night Count	24hr Count	Day Length	Night Length	24hr Length
GAARPALA	0.0	0	No Sound Audible	0.0	No Sound Audible	0.0	0.0	0.0	0.0	0.0	0.0
GAARPALA	1.1	1	Aircraft	1.1	Jet	1.2	1.1	2.3	119.3	135.2	126.8
GAARPALA	1.2	1	Aircraft	1.2	Prop	2.0	0.1	2.1	167.6	329.3	173.6

Table 8. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2019.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	3600	3600	3600	3600	3600	3600	384	3600	1135	128	930	1486	1062	1546	580	1584	3600	1074	1348	3600	3600	3600	1024	GAARITKI
50%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	2474	3600	3600	3600	3600	3600	3600	3600	3600	GAARITKI
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARITKI
Mean	3600	3600	3600	3600	3600	3600	3600	2460	3600	2989	2126	2474	3150	2548	3149	1847	3172	3600	2655	3087	3600	3600	3600	2431	GAARITKI

Table 9. SPLAT: duration of noise−free intervals (seconds) between detected events, at the 90th/50th/10th percentile and mean, by hour of day, in Summer 2019.

Percentile	00h	01h	02h	03h	04h	05h	06h	07h	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h	SiteID
90%	3600	3600	3600	1088	373	1249	2284	3391	1734	1451	882	504	1640	1458	822	476	911	706	1528	2490	3530	3600	3600	3600	GAARPALA
50%	3600	3600	3600	3600	3014	3600	3600	3600	3600	3600	3600	3551	3600	3600	3600	3600	3600	1911	3600	3600	3600	3600	3600	3600	GAARPALA
10%	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	3600	GAARPALA
Mean	3600	3600	3360	2817	1785	2700	3213	3191	3036	2831	2255	2049	2984	2963	2252	2165	2712	1674	2833	3130	2964	3232	3451	3600	GAARPALA

Table 10. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2019.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARITKI	No Sound Audible	0	0	0	0	0	0	0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0	0.0	0.0	0	0	0	0.0
GAARITKI	Jet	0	0	0	0	0	0	0	0.3	0	0.0	0.6	0.5	0.5	1.0	0.0	1.5	0.0	0	0.4	0.0	0	0	0	0.0
GAARITKI	Prop	0	0	0	0	0	0	0	0.8	0	1.4	0.5	0.3	0.0	0.0	0.9	0.0	0.4	0	0.0	0.2	0	0	0	0.6
GAARITKI	Helicopter	0	0	0	0	0	0	0	0.0	0	0.0	0.0	0.0	0.0	0.3	0.0	0.2	0.0	0	0.0	0.0	0	0	0	0.0

Table 11. SPLAT (automated detection): percent of time each individual sound source was detected, by hour of day, in Summer 2019.

SiteID	Source Description	00h	01h	02h	03h	04h	05h	06hr	07hr	08h	09h	10h	11h	12h	13h	14h	15h	16h	17h	18h	19h	20h	21h	22h	23h
GAARPALA	No Sound Audible	0	0	0.0	0.0	0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0
GAARPALA	Jet	0	0	0.0	0.6	2	0.8	0.3	0.2	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.4	0.2	2.2	0.4	0.3	0.0	0.1	0.0	0
GAARPALA	Prop	0	0	0.2	0.0	0	0.0	0.0	0.0	0.3	0.8	1.3	2.0	0.4	0.8	1.2	1.1	0.8	0.5	0.0	0.0	0.2	0.0	0.3	0

Table 12. Percent time above sound level thresholds for acoustic records collected at Gates of the Arctic National Park and Preserve (GAAR) in Summer, 2019. Values show the percent of time ANS−weighted sound levels (20−1,250 Hz) and the full measured frequency range (12.5−20,000 Hz) exceeded 35, 45, 52, and 60 dB (LAeq,1s), by site and day/night.

Site ID	Frequency (Hz)	35 dB Day	45 dB Day	52 dB Day	60 dB Day	35 dB Night	45 dB Night	52 dB Night	60 dB Night
GAARITKI	Full (12.5−20,000)	7.5	0.6	0.1	0	10.0	0.7	0.1	0
GAARITKI	ANS (20−1,250)	2.1	0.0	0.0	0	4.9	0.0	0.0	0
GAARPALA	Full (12.5−20,000)	2.2	0.4	0.0	0	2.0	0.2	0.0	0
GAARPALA	ANS (20−1,250)	0.3	0.0	0.0	0	0.2	0.0	0.0	0

Figure 1. GAARITKI: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2019.

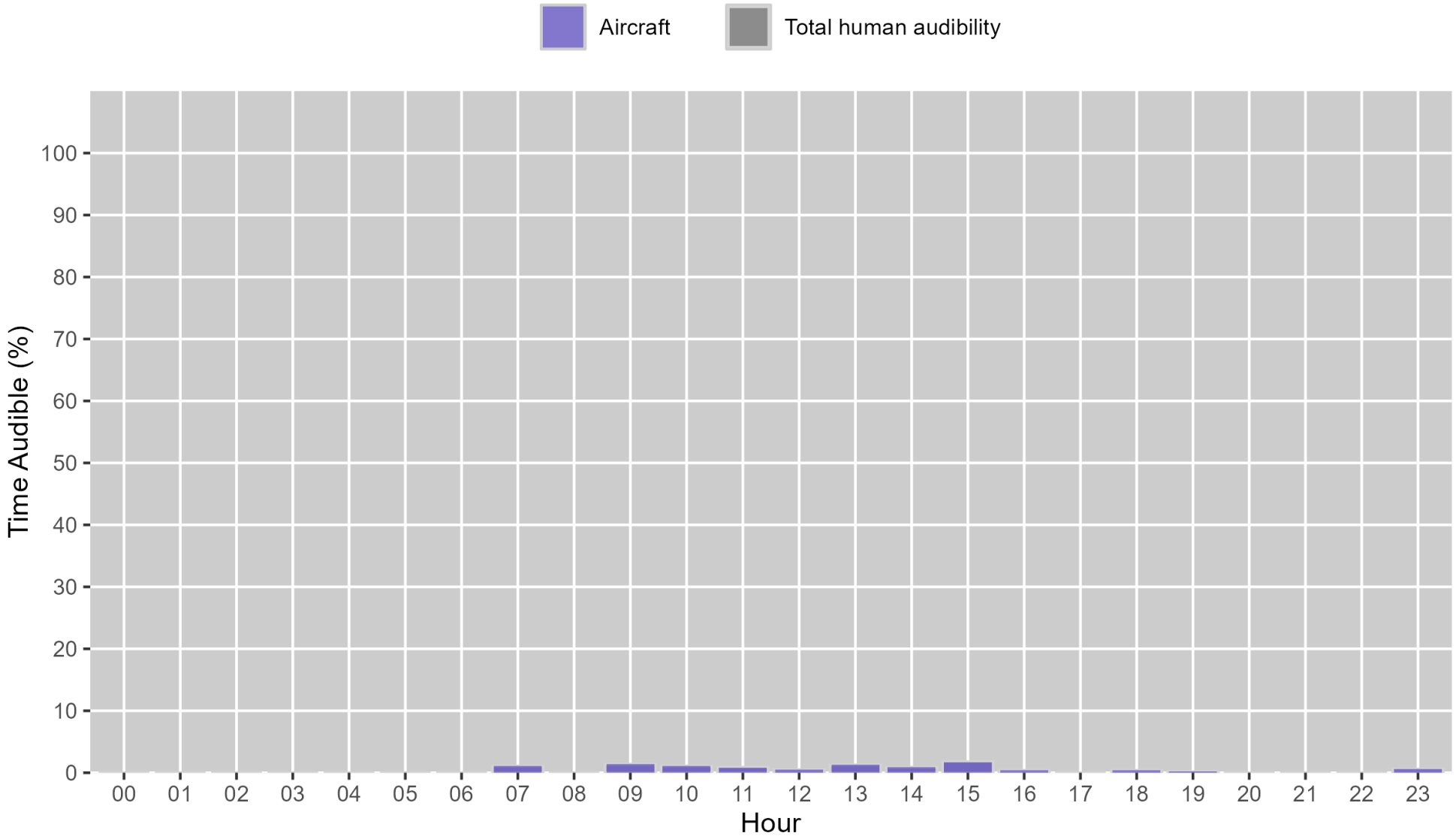


Figure 2. GAARITKI: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2019.

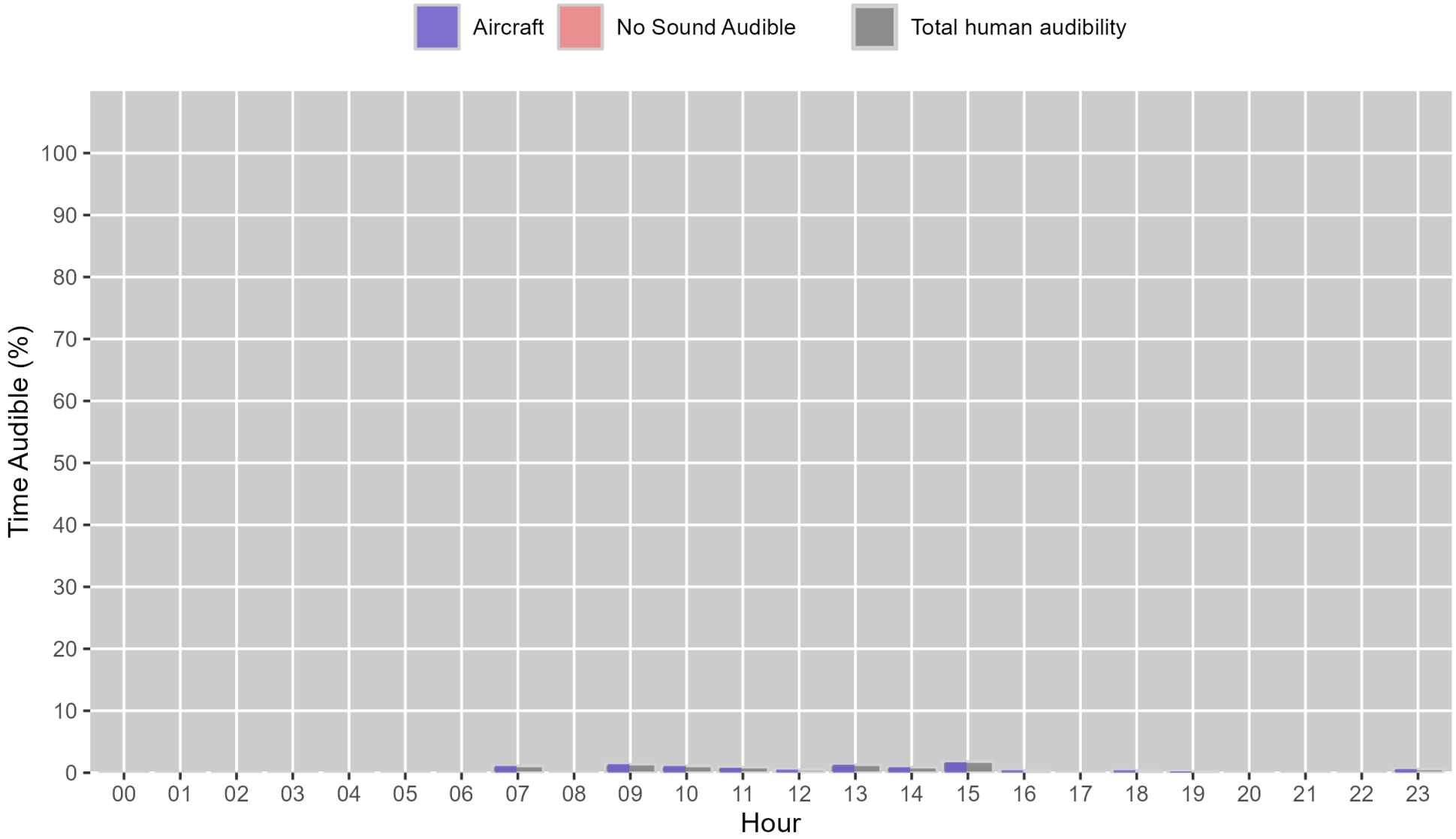


Figure 3. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARITKI in Summer, 2019, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

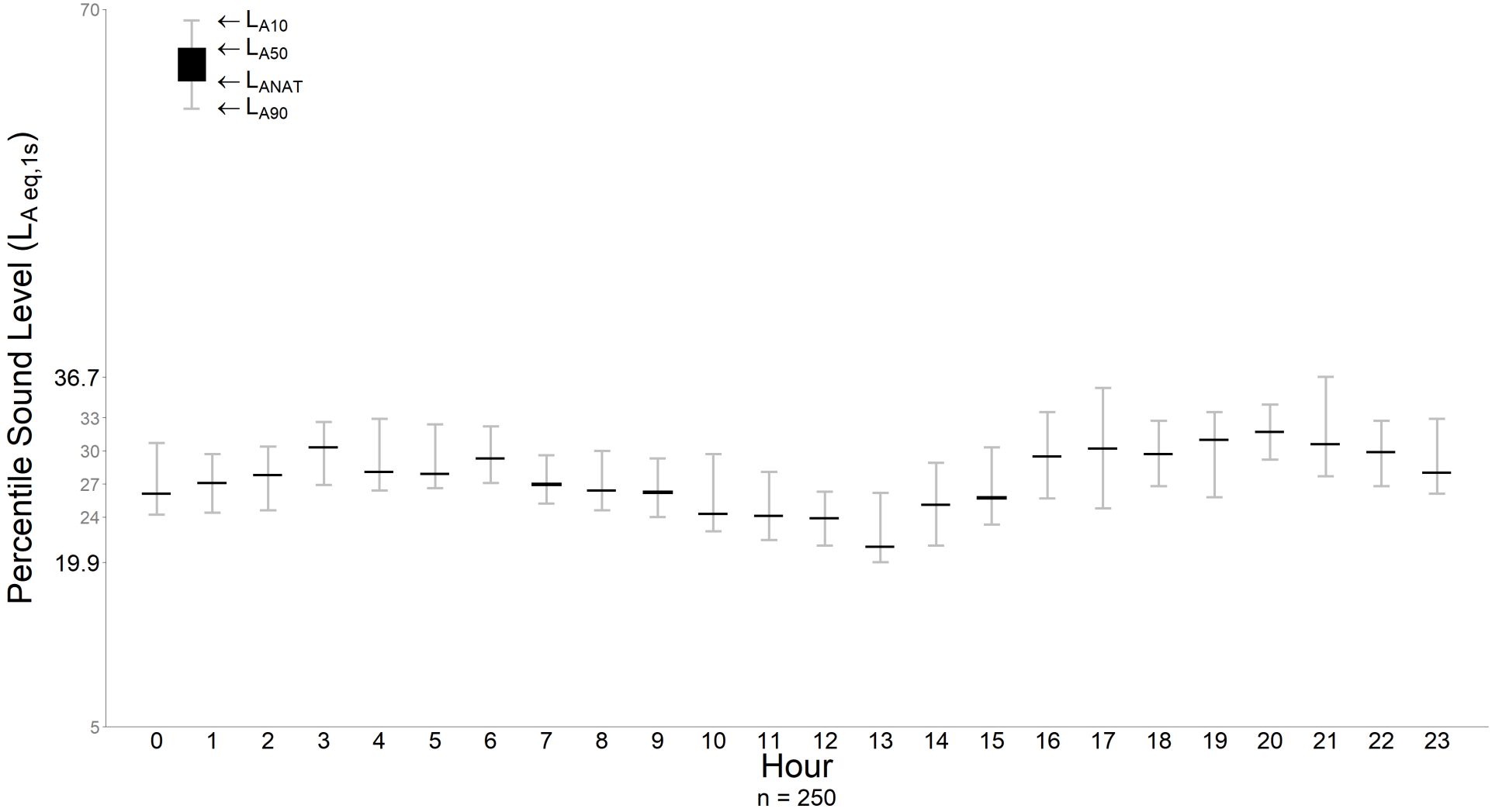


Figure 4. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARITKI in Summer, 2019.

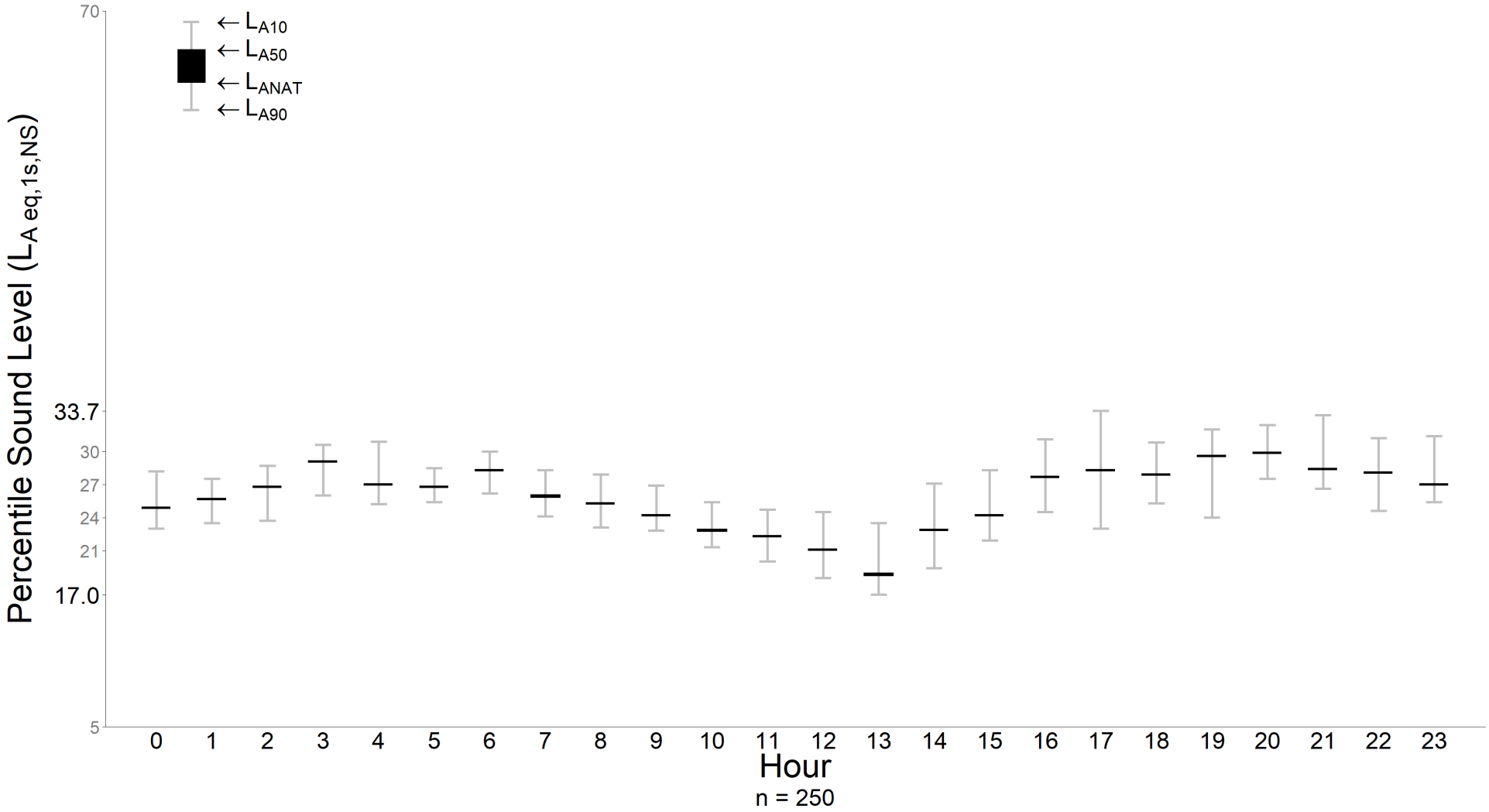


Figure 5. GAARITKI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

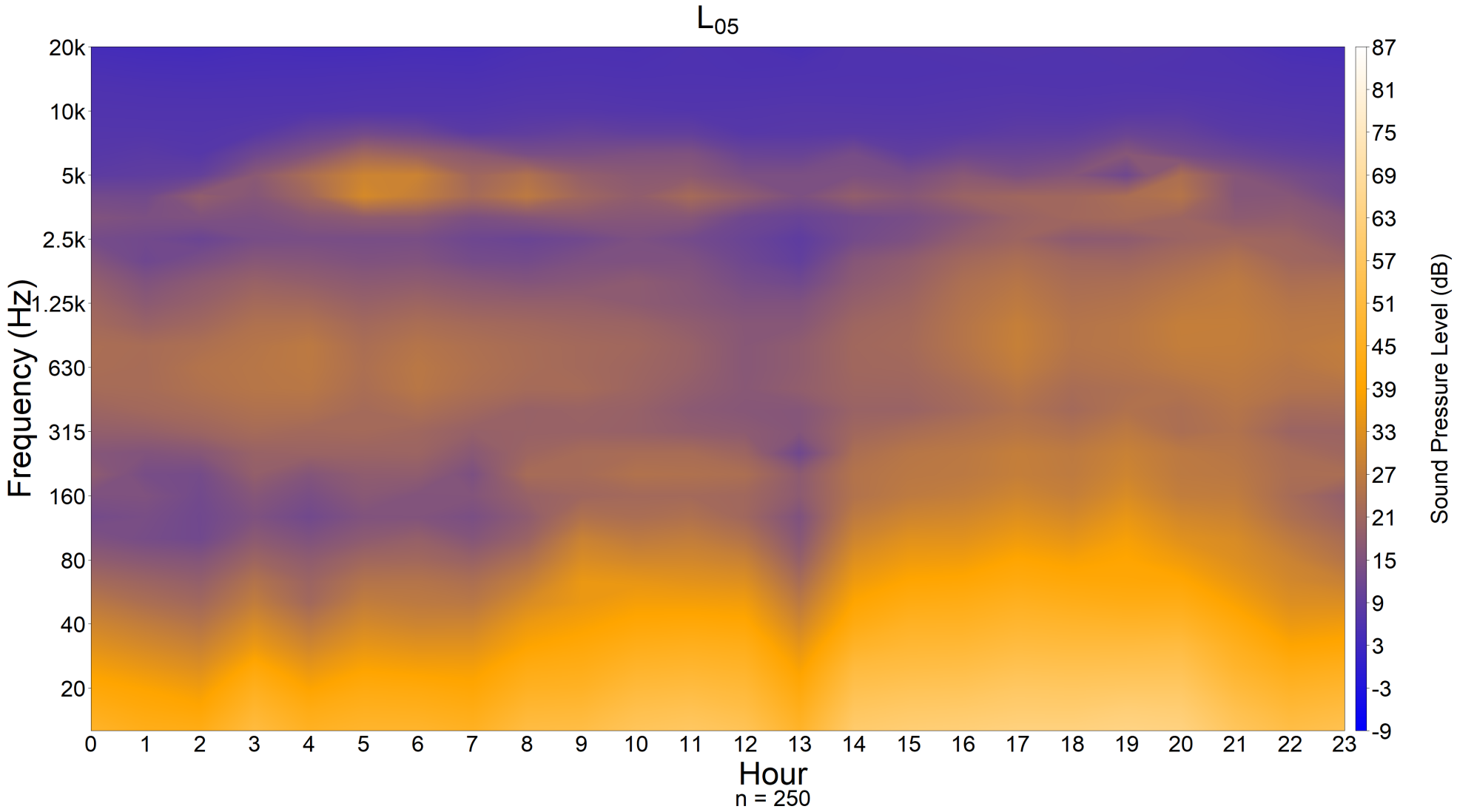


Figure 6. GAARITKI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

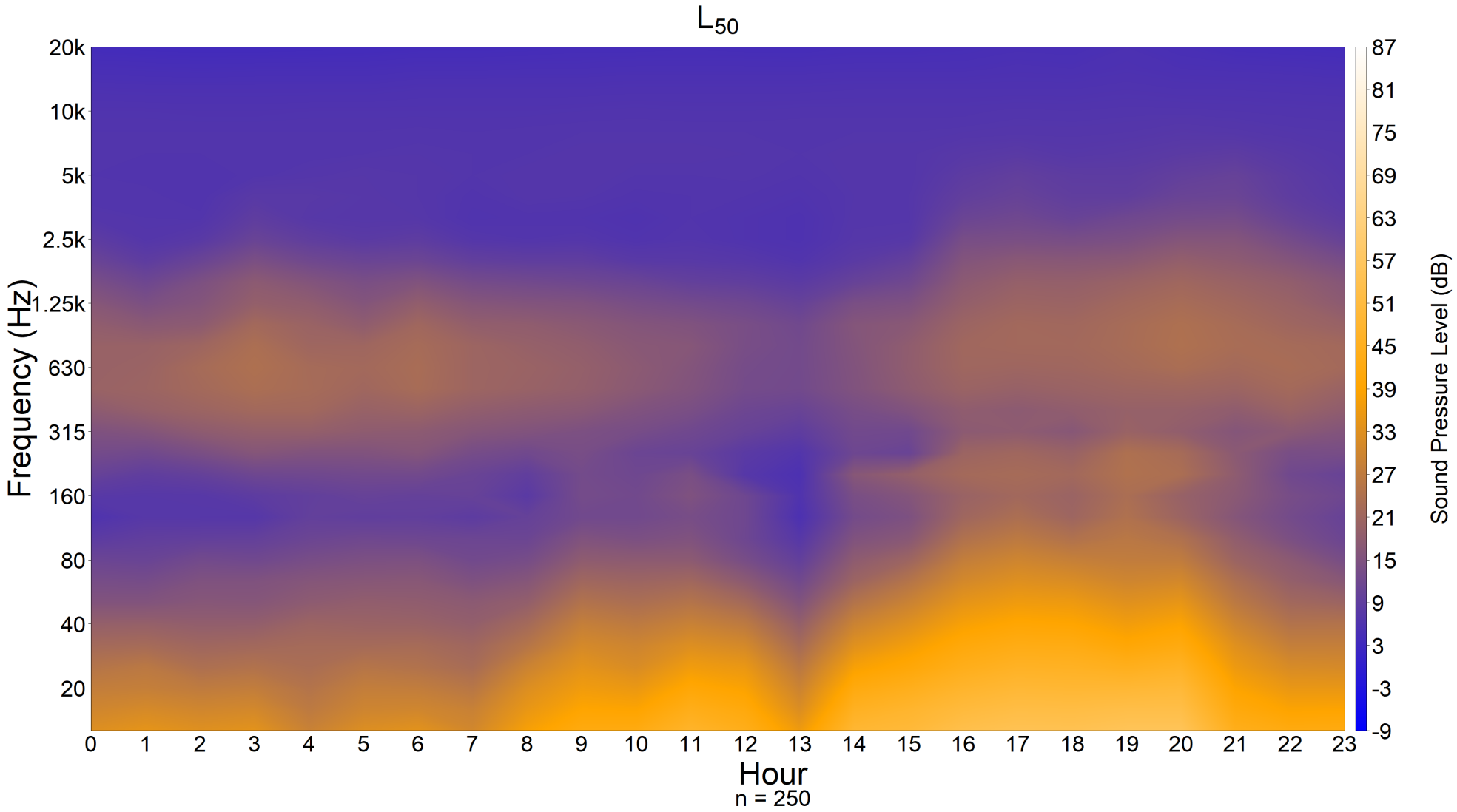


Figure 7. GAARITKI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

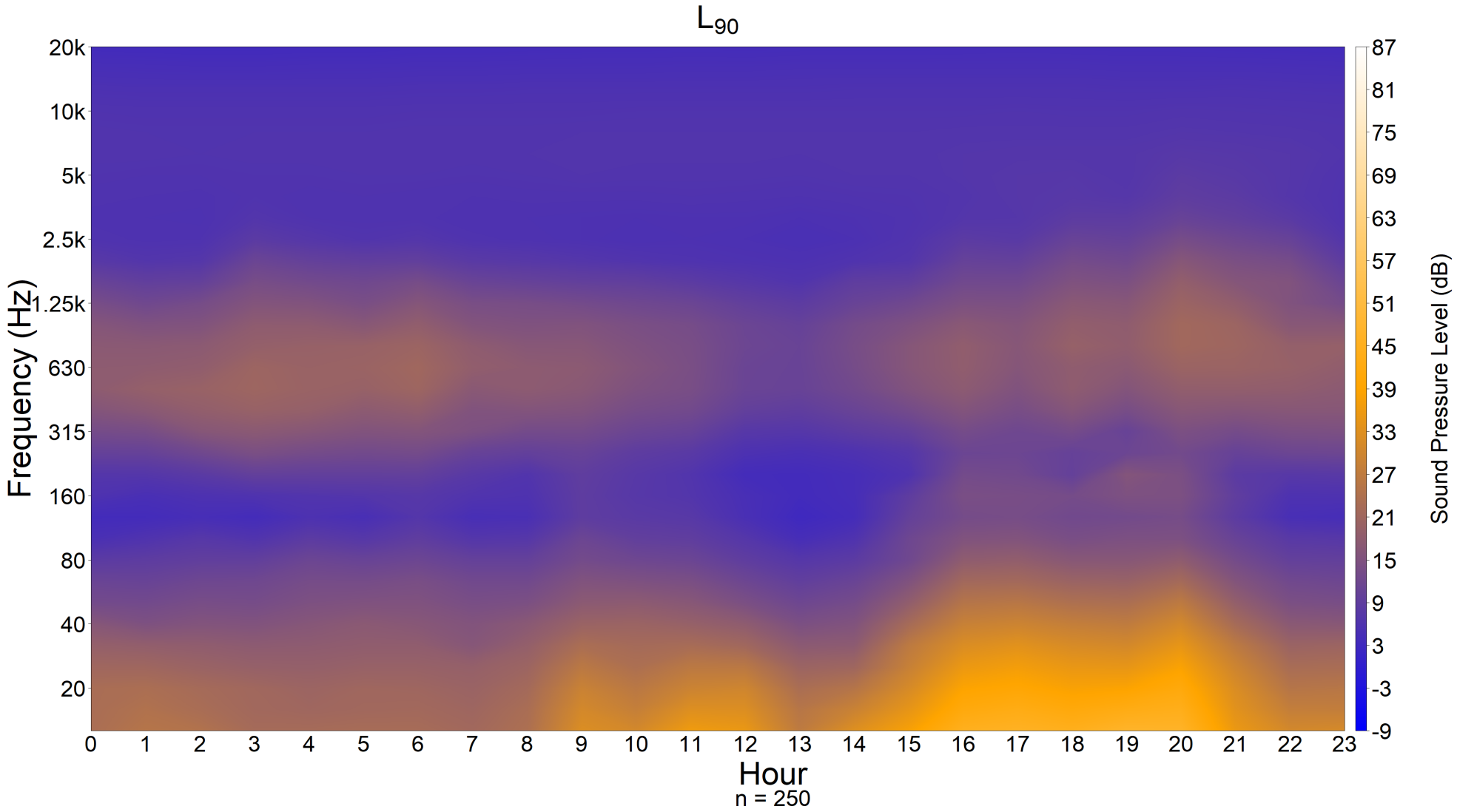


Figure 8. GAARITKI: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

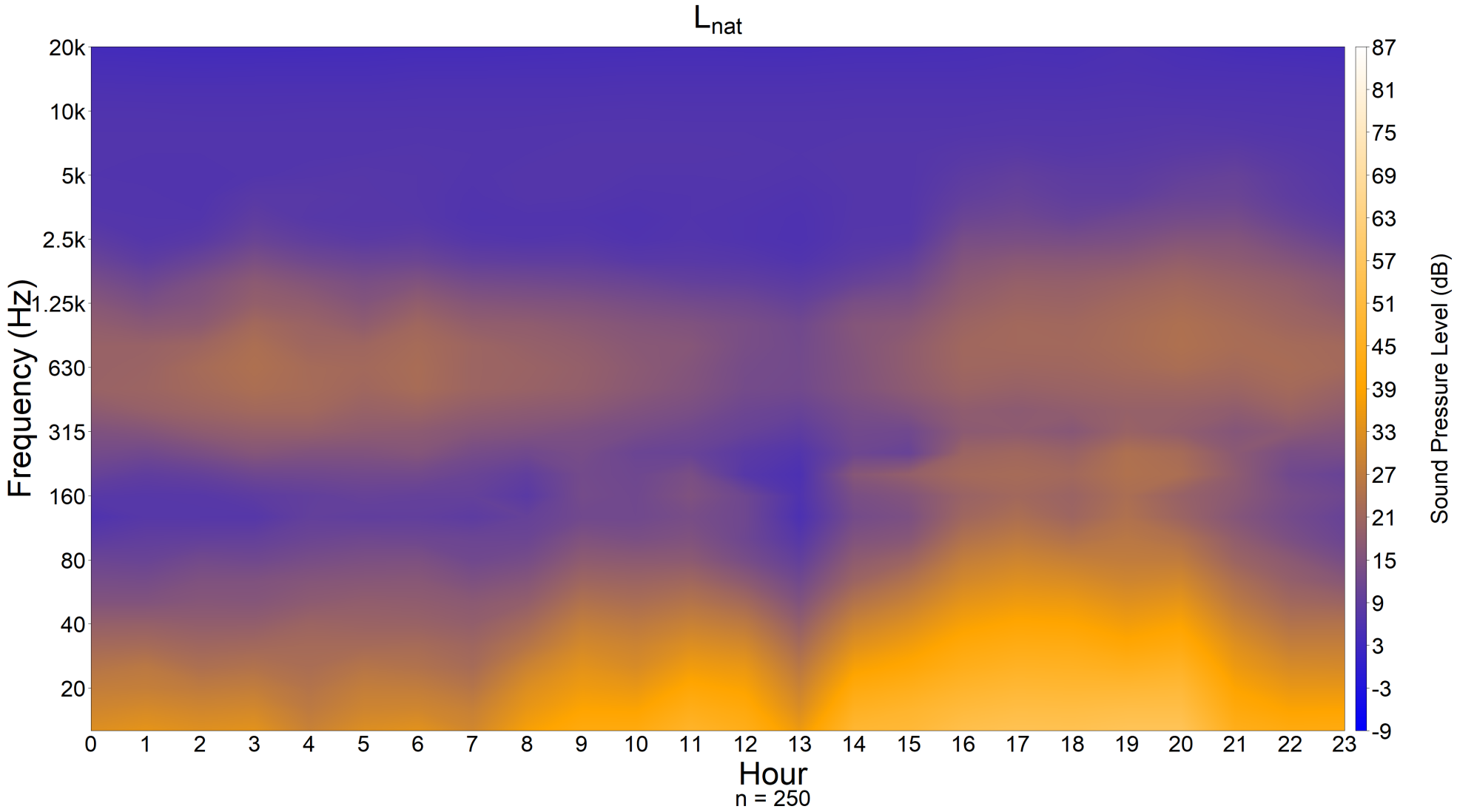


Figure 9. GAARITKI: median and mean duration of noise−free intervals by hour of day, in Summer 2019.

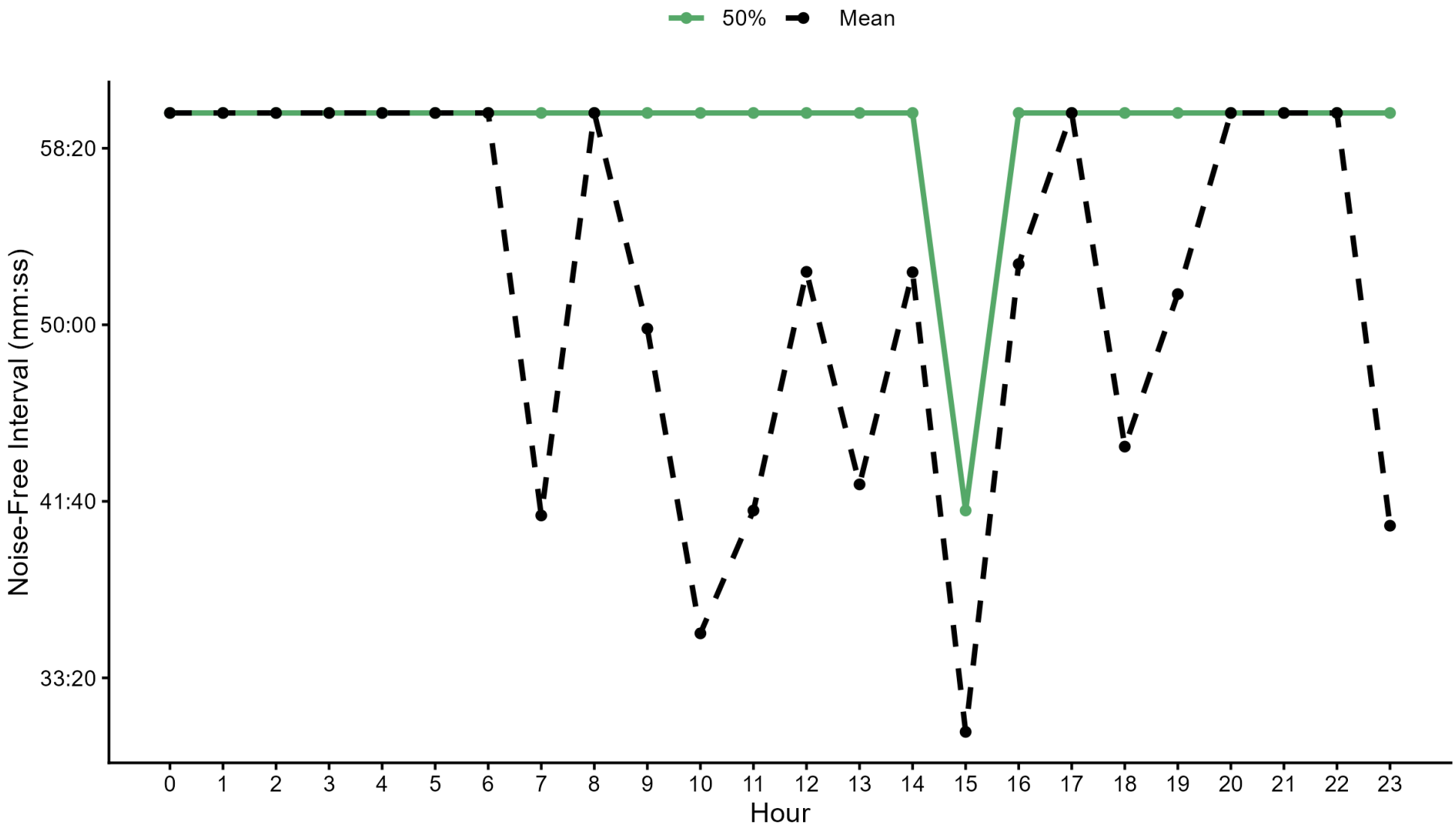


Figure 10. GAARITKI: percent of time audible by sound source category (noise/natural), in Summer 2019.

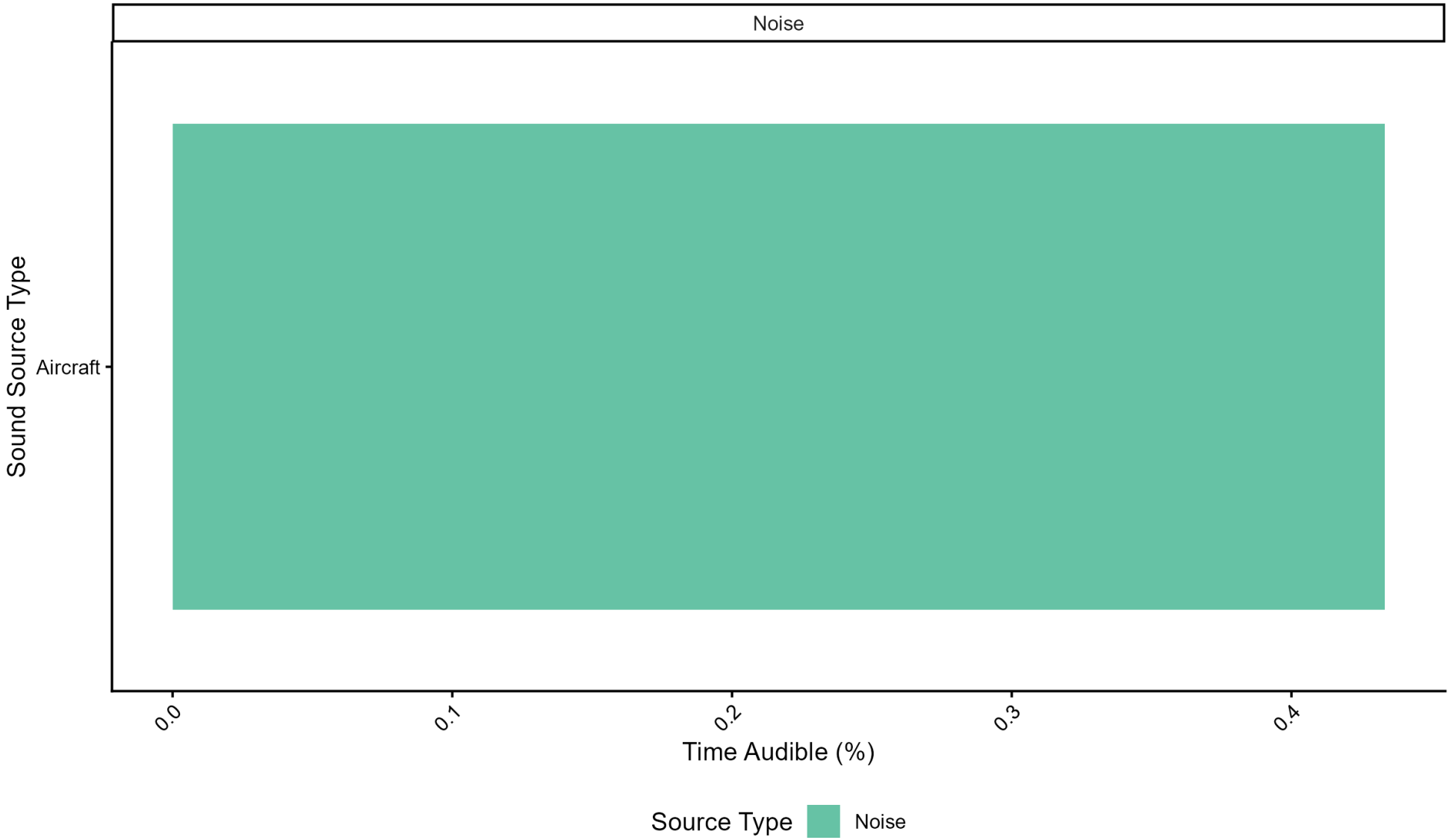


Figure 11. GAARITKI: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2019.

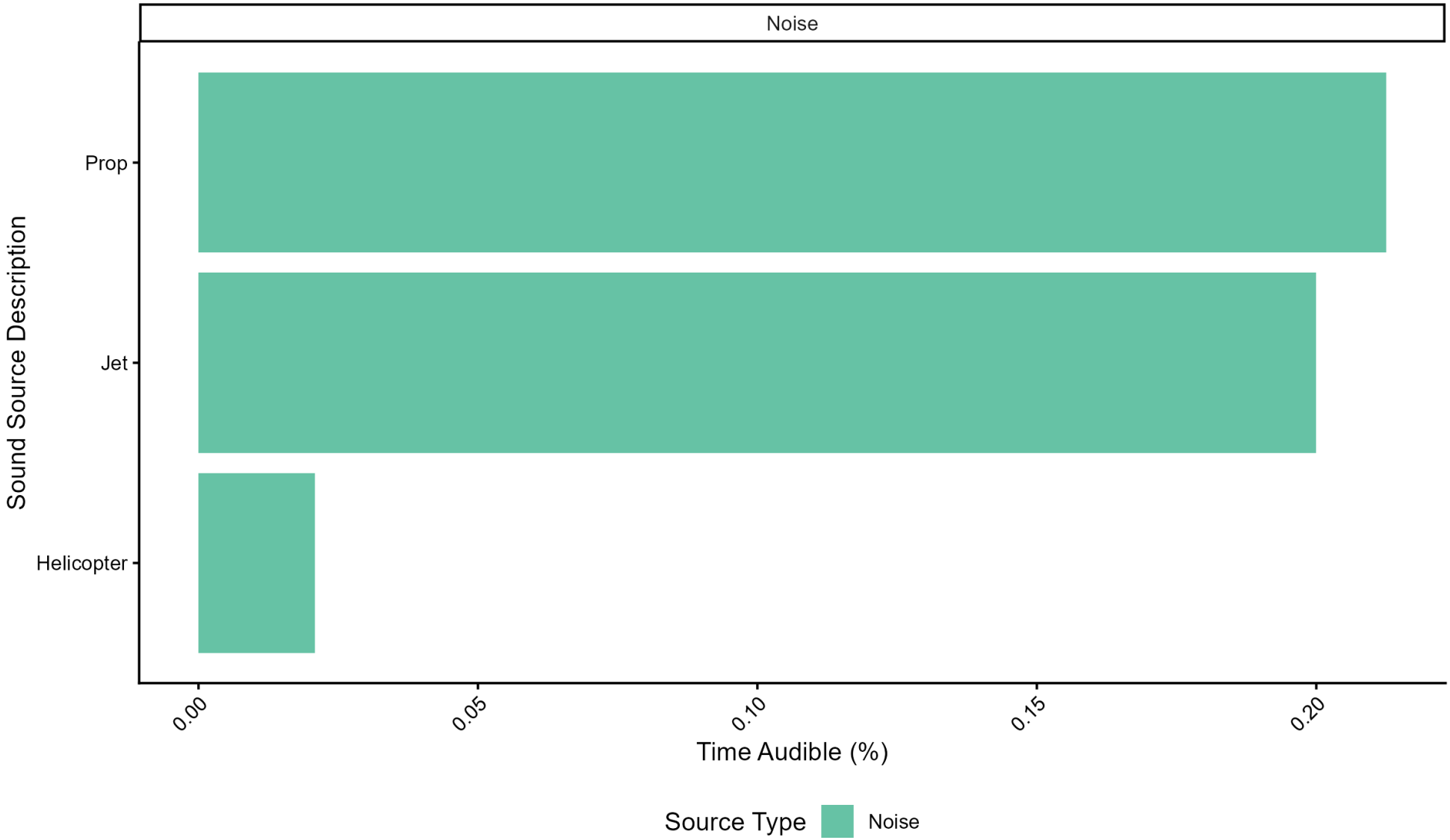


Figure 12. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARITKI in Summer, 2019. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

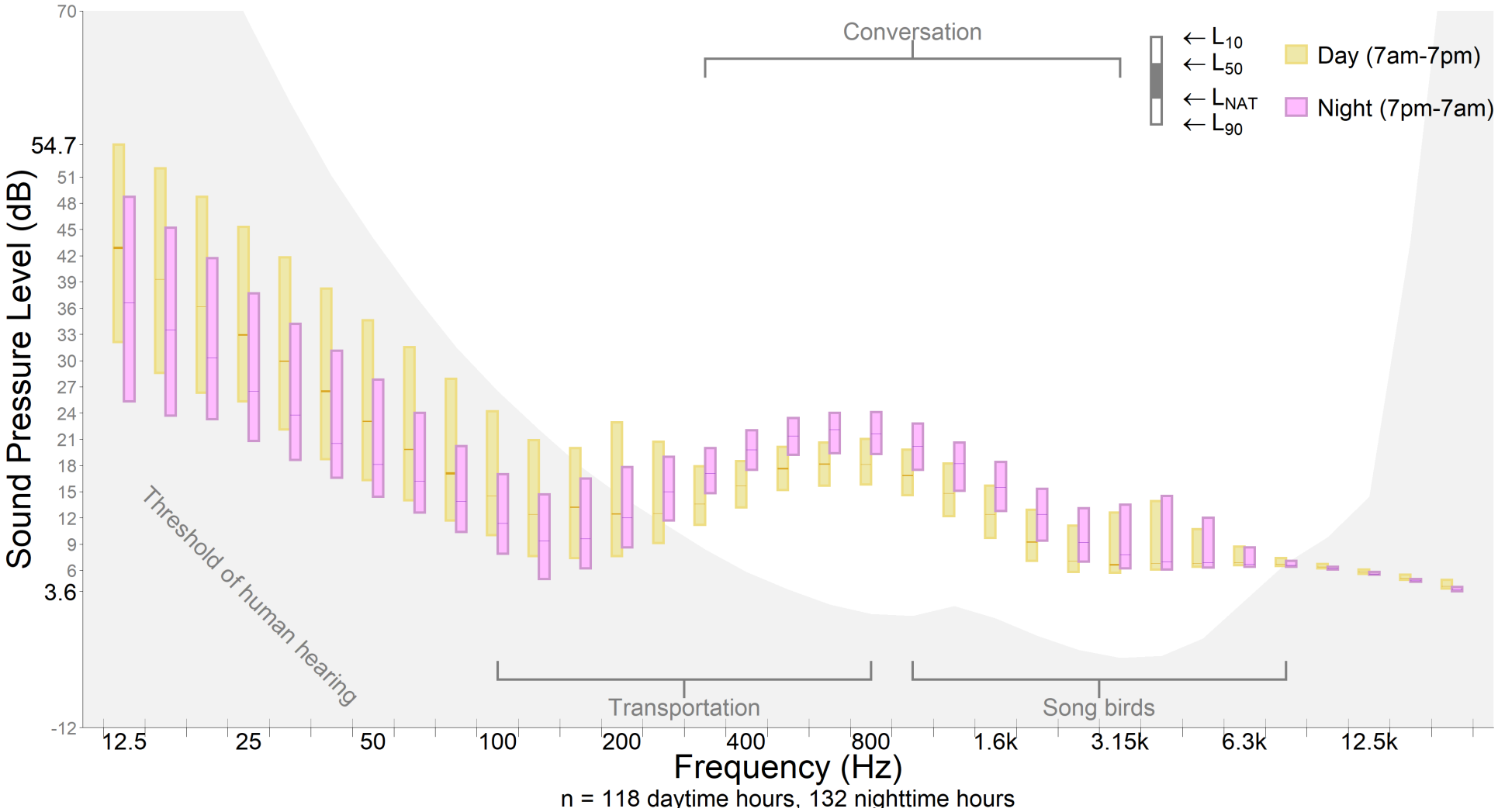


Figure 13. GAARPALA: hourly percent time audible for total human−caused sound compared to the single top noise source, in Summer 2019.

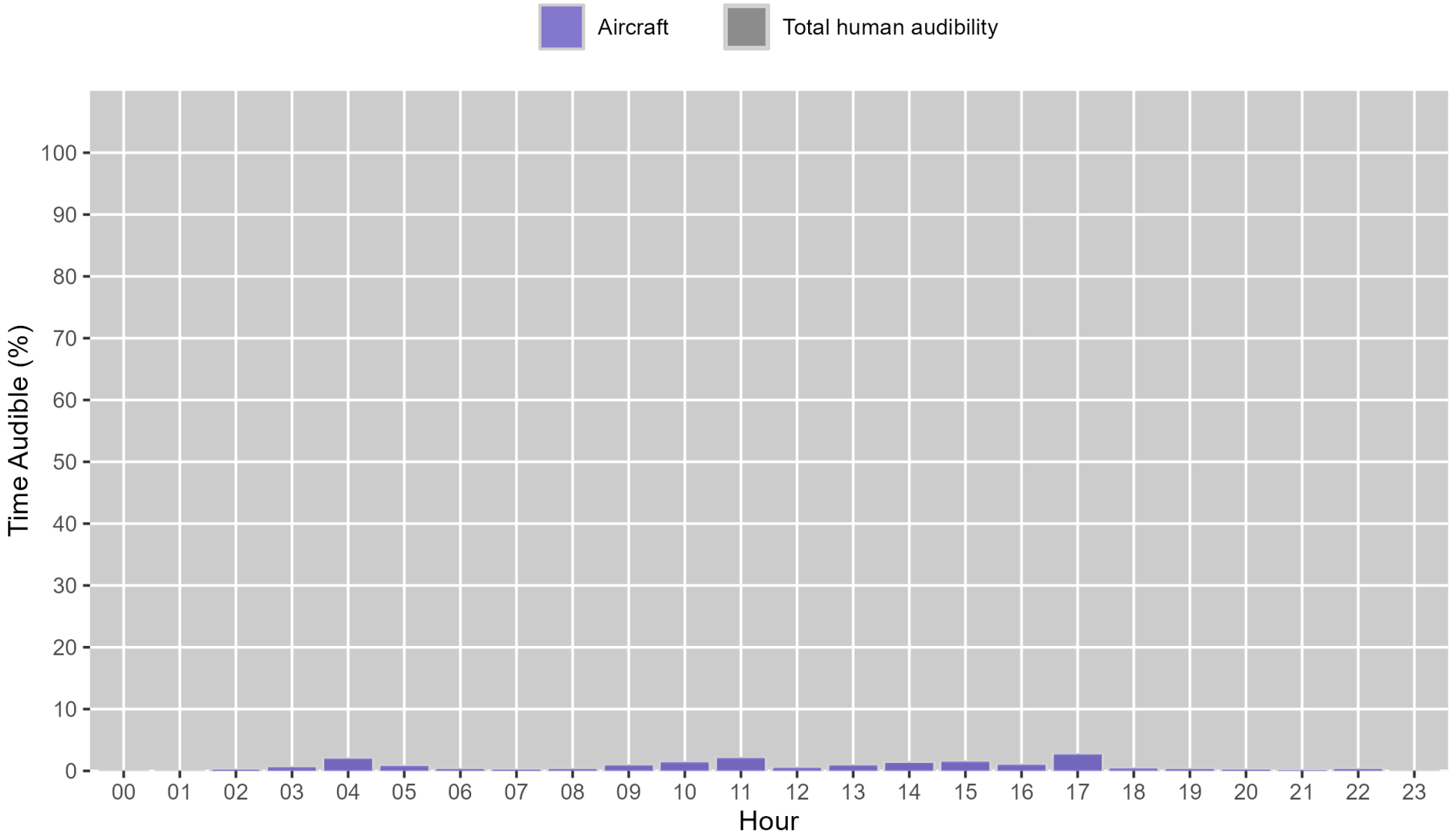


Figure 14. GAARPALA: hourly percent time audible for total human−caused sound compared to the top two noise sources, in Summer 2019.

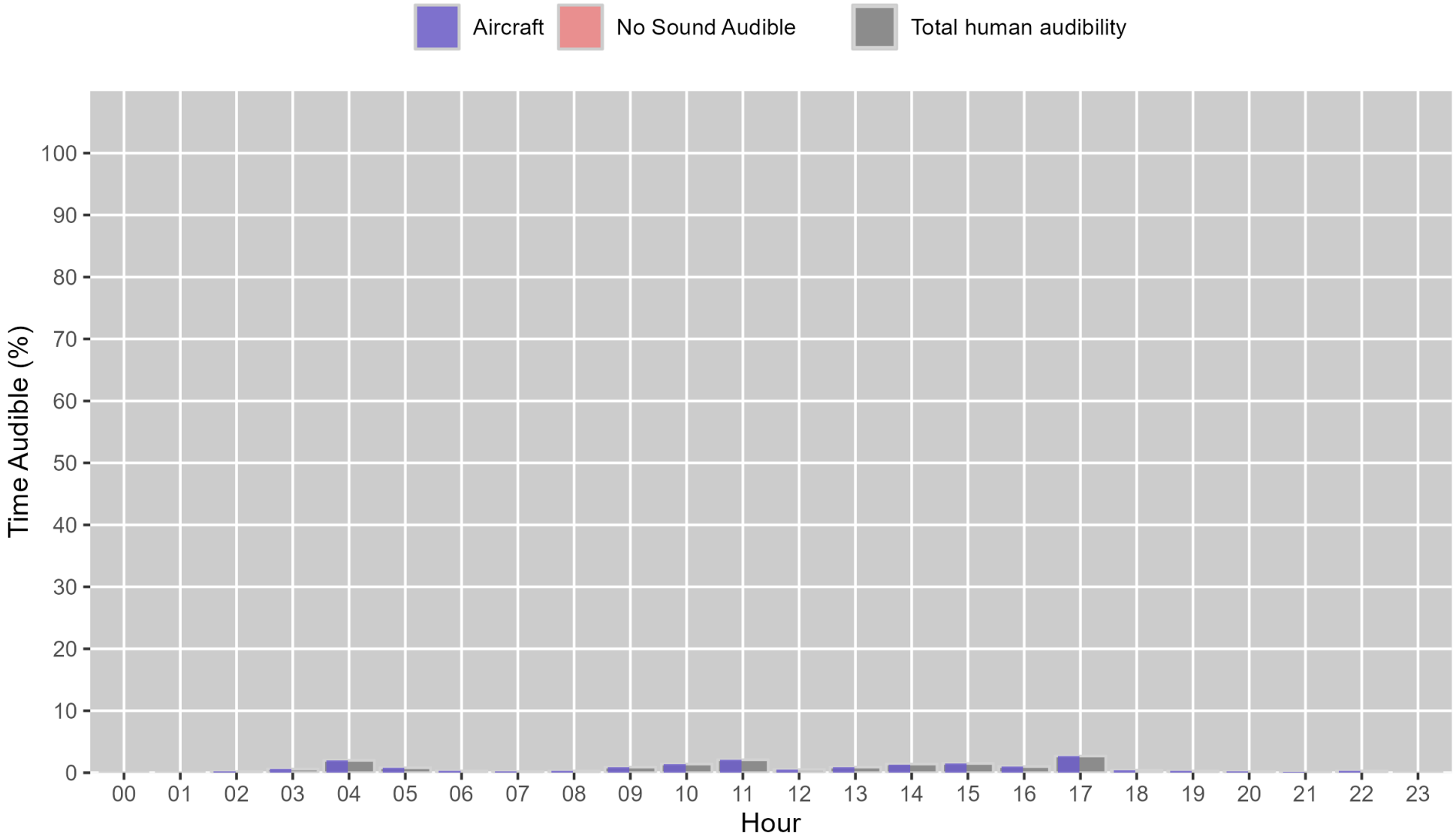


Figure 15. Hourly percentile sound levels (LAeq,1s, dB re 20 uPa) at GAARPALA in Summer, 2019, showing LA10, LA50, LA90 and (when available) LAnat for each hour of the day.

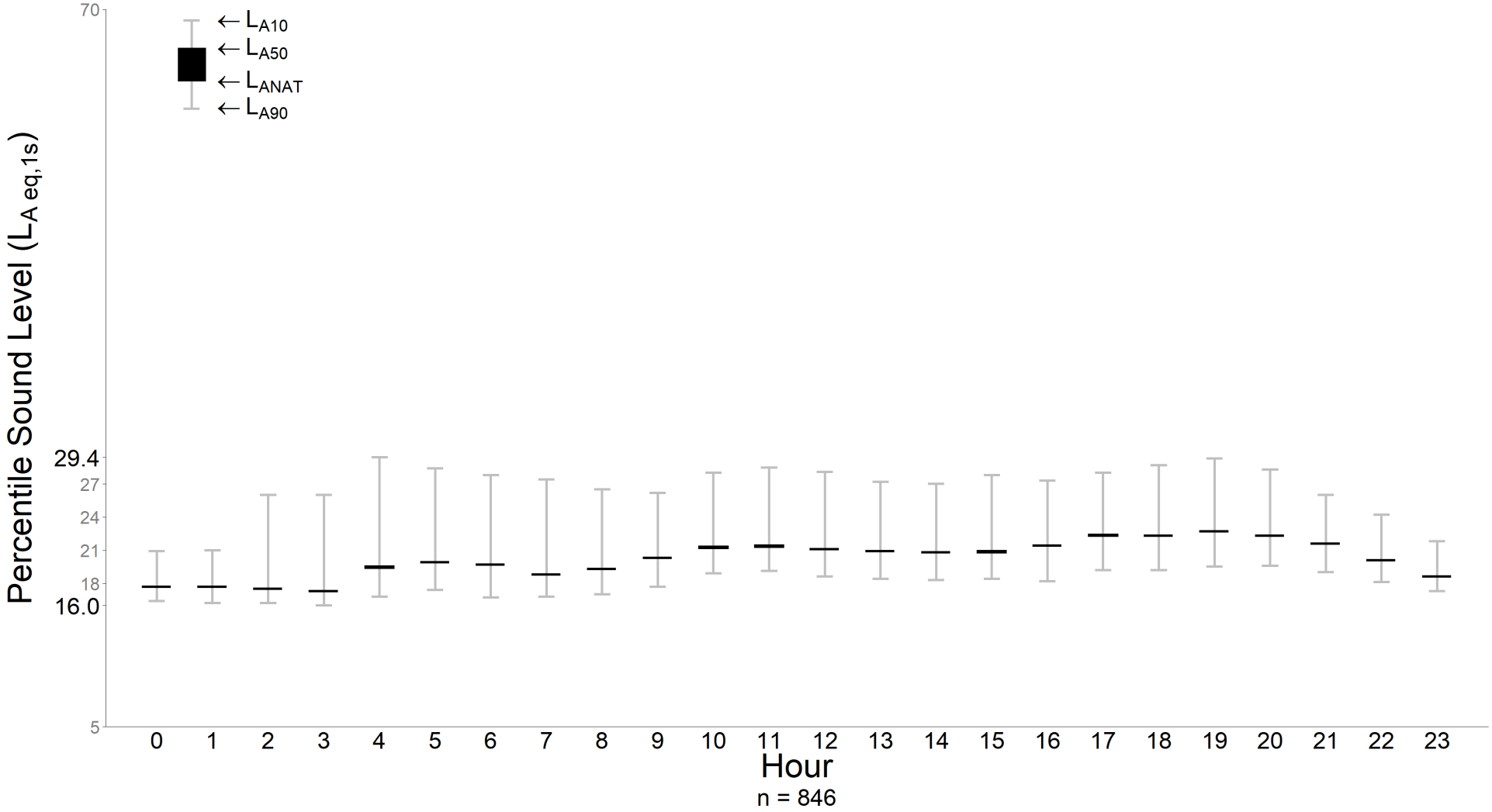


Figure 16. Hourly percentile sound levels, ANS−weighted (20−1,250 Hz), at GAARPALA in Summer, 2019.

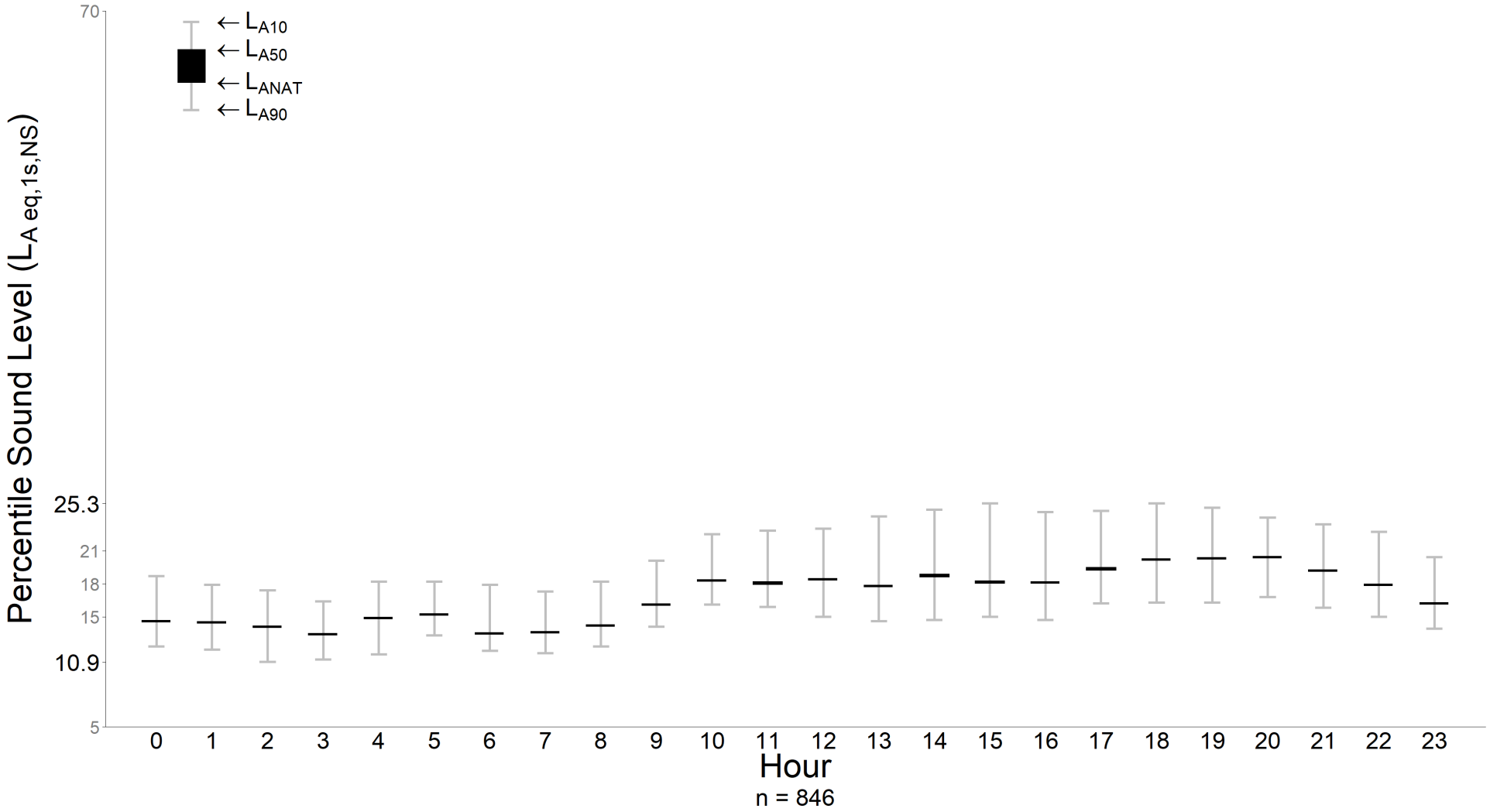


Figure 17. GAARPALA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

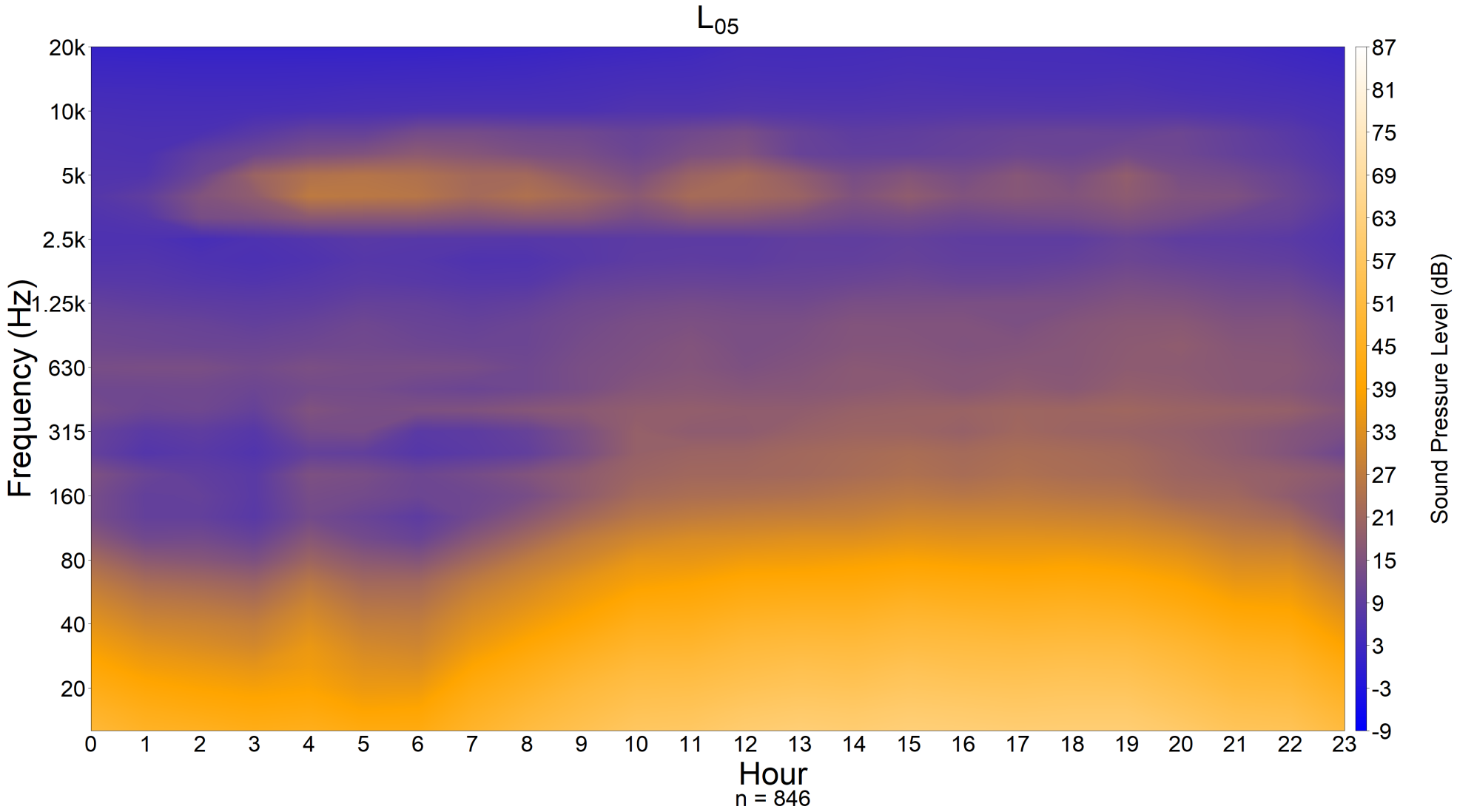


Figure 18. GAARPALA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

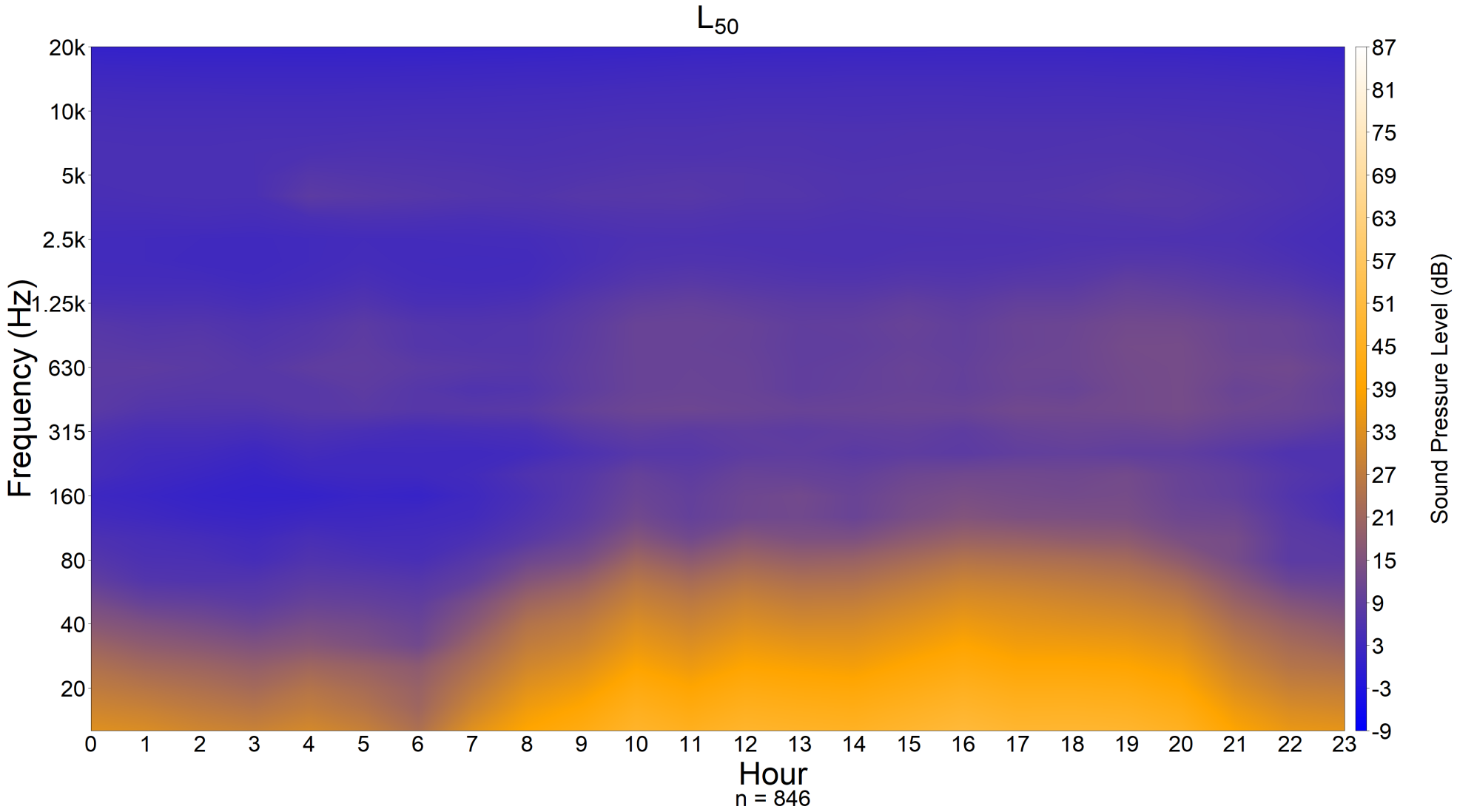


Figure 19. GAARPALA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

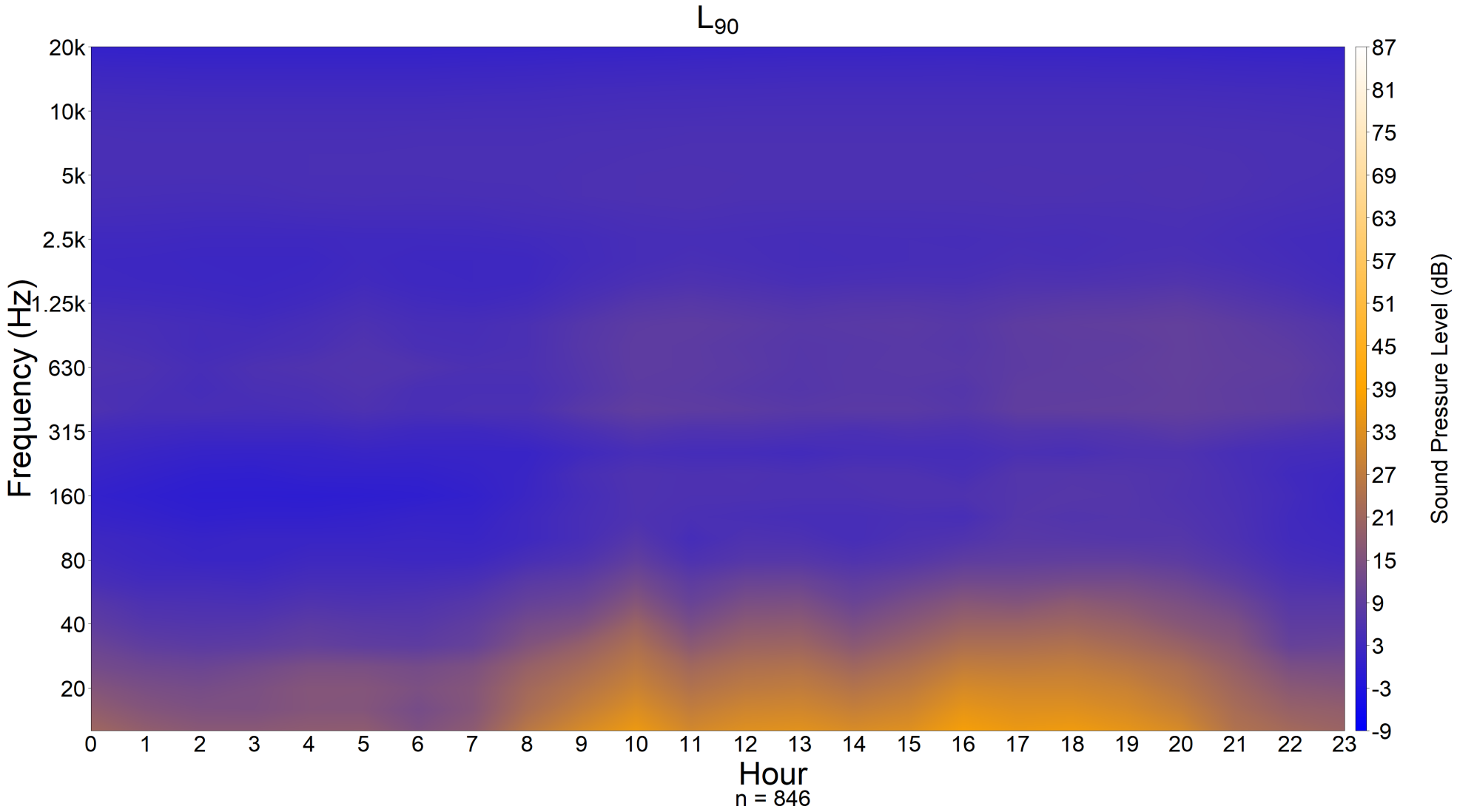


Figure 20. GAARPALA: sound level contour plot (hour of day vs. frequency, colored by sound pressure level in dB), Summer 2019.

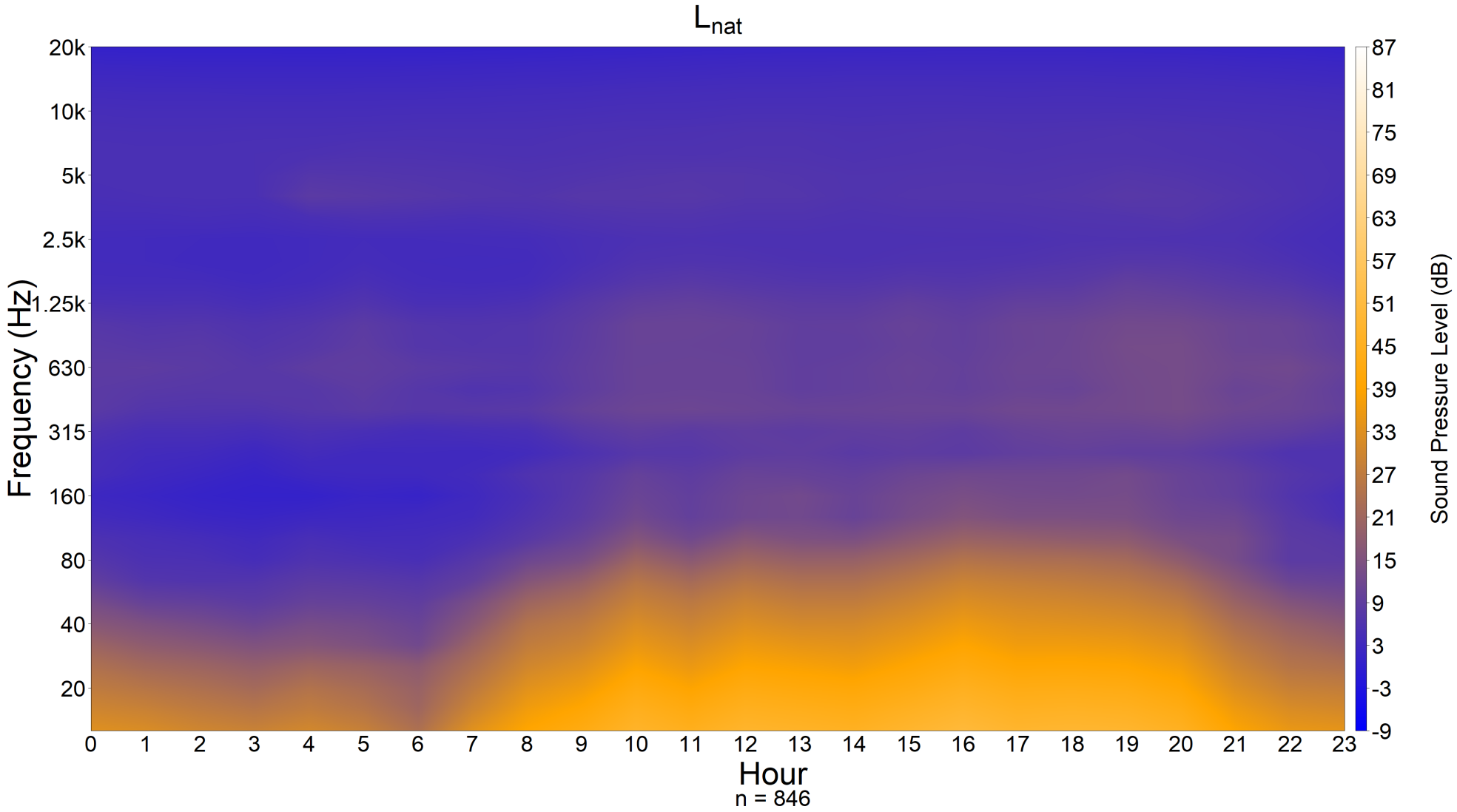


Figure 21. GAARPALA: median and mean duration of noise−free intervals by hour of day, in Summer 2019.

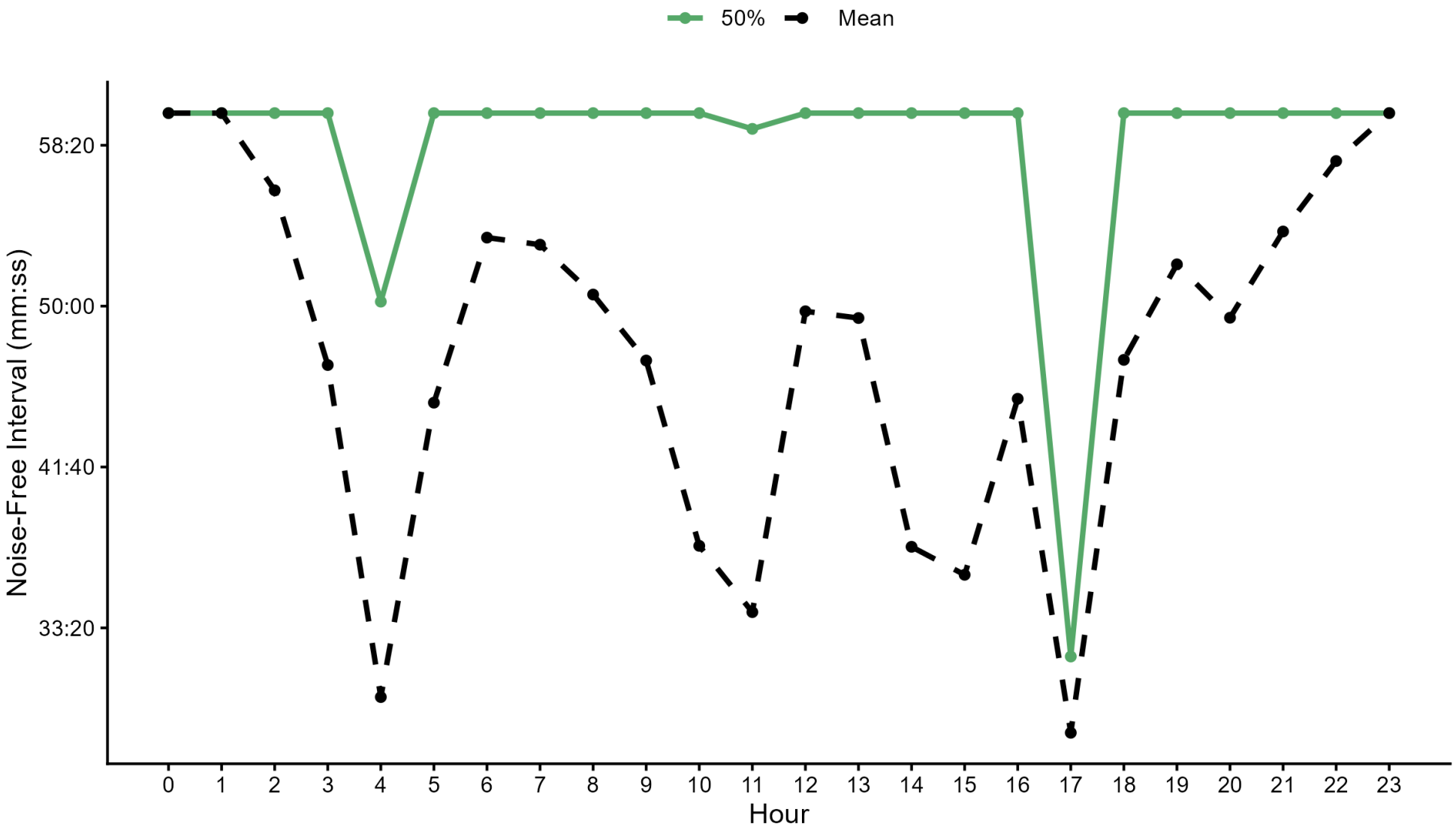


Figure 22. GAARPALA: percent of time audible by sound source category (noise/natural), in Summer 2019.

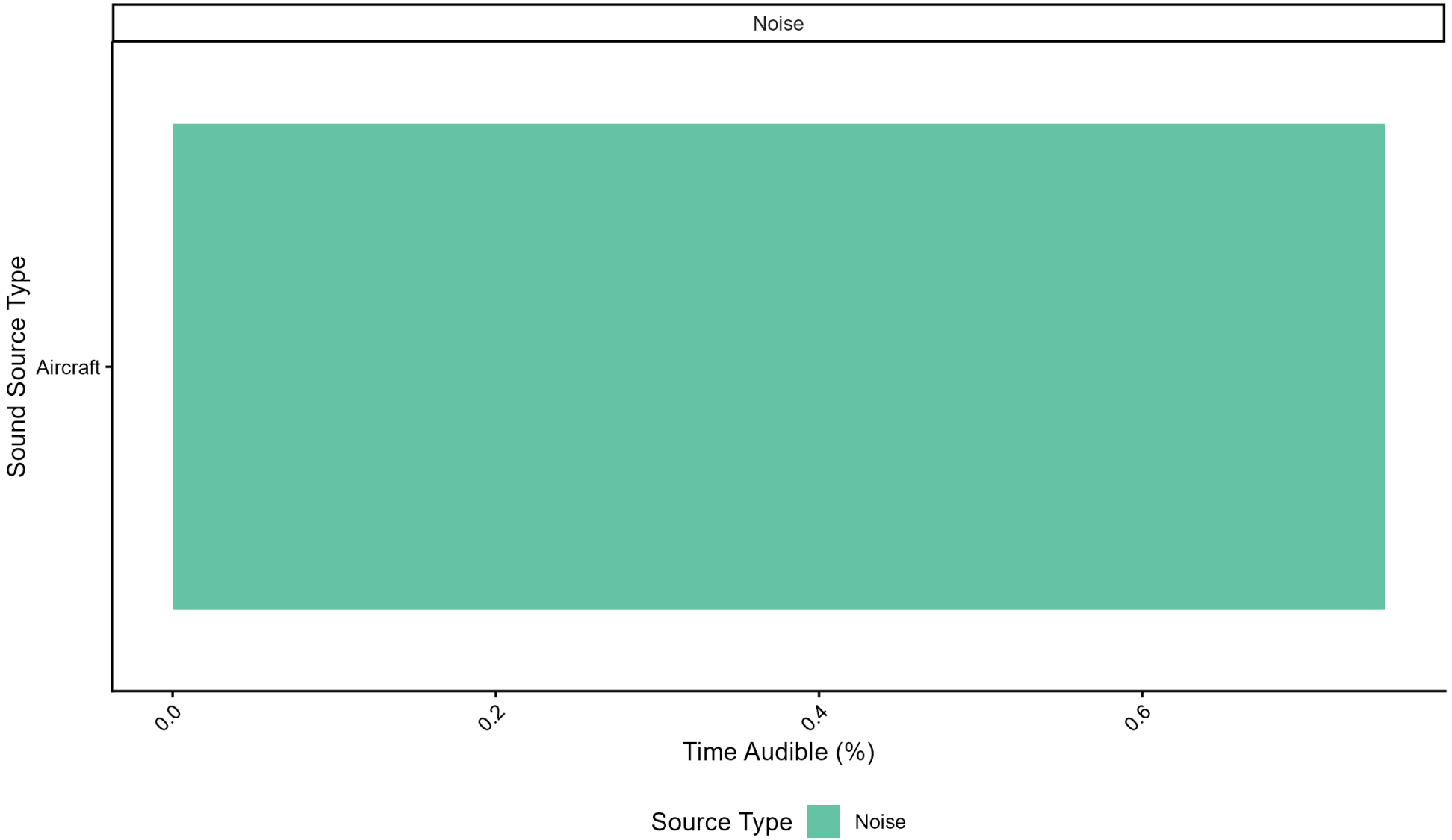


Figure 23. GAARPALA: percent of time each individual sound source was audible, grouped by category (noise/natural), in Summer 2019.

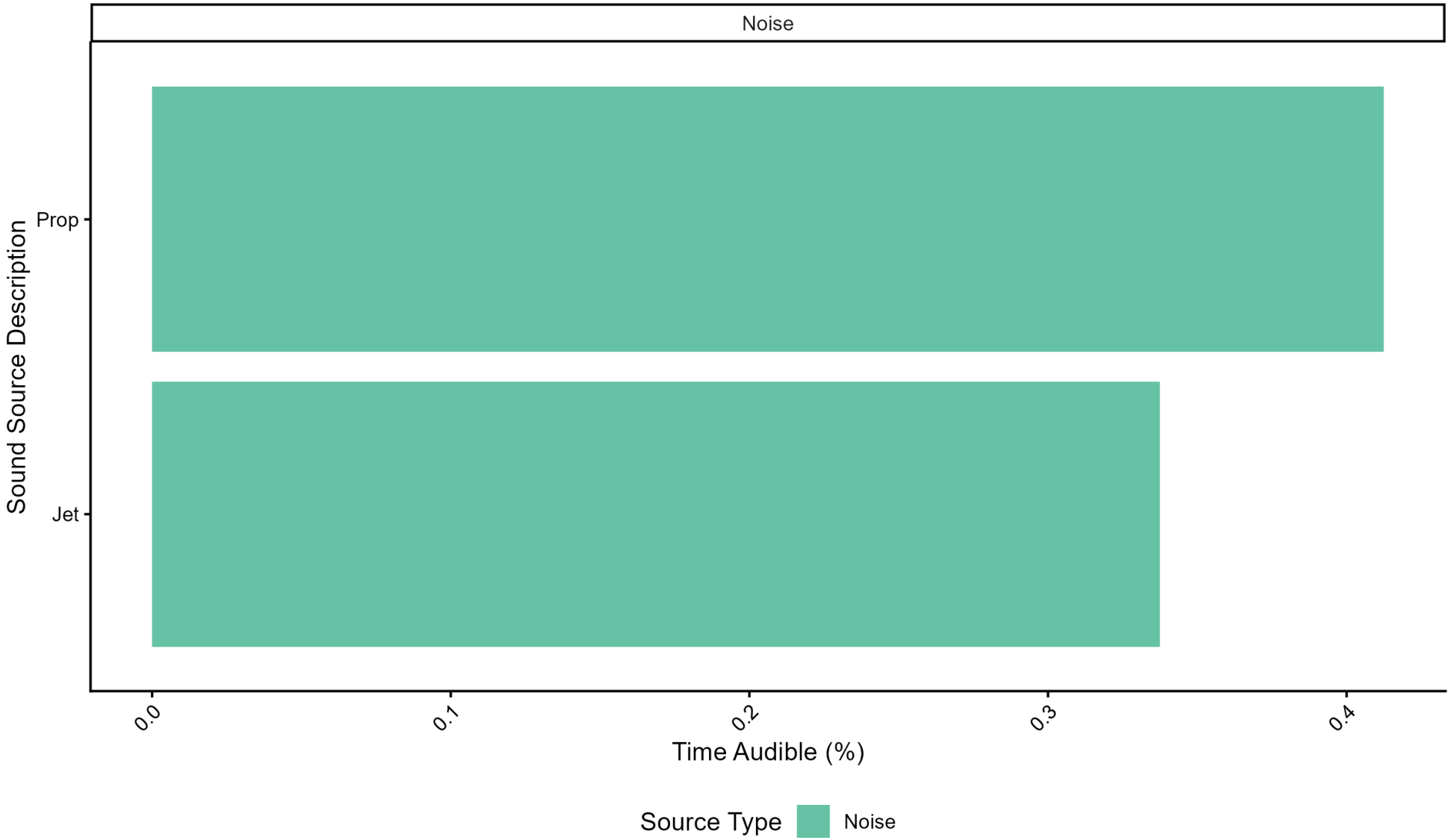


Figure 24. Day and night percentile sound pressure levels for 33 one−third octave bands at GAARPALA in Summer, 2019. The bold portion of each bar indicates the range between existing (L50) and natural (Lnat) sound pressure levels; grayed areas fall outside the typical range of human hearing.

